Final Presentation

Comparing apples and pears? Impacts of using the Jacobian scaled conditional Akaike Information Criteria for model selection in linear mixed models. Advanced Small Area Estimation (Prof. Dr. Schmid)

Niklas Ippisch

2026-07-22

Table of Contents I

Motivation

Motivation

- Situation: Linear Mixed Model with $y_i = X_i\beta + Z_i u_i + \epsilon_i$
- Adequate set of variables essential for good model fit
- Data driven transformation can improve the model fit
- Transformations especially important if normality assumption(s) $u_i \sim N(0, G)$ and/or $\epsilon_i \sim N(0, \sigma^2 I_n)$ not fulfilled
- Problem: Variable selection and estimation of transformation parameter λ affect each other:
  - Optimal set of covariates depends on transformation chosen and vice versa
  - Other papers suggest the usage of a conditional Akaike Information Criteria: $cAIC = -2log(g(y|\hat{\theta}, \hat{u})) + 2K_{MLE}$
  - Problem: Not scale invariant
  - Naive solution: variable selection on original scale and transformation afterwards

Proposed solution

Proposed Solution by Lee et al. (2023) I

- Idea: Change of variable approach: $l(y|\theta, u) = g(\tilde{y}|\theta, u) \cdot J(\lambda, y)$, with $l(y|\theta, u)$ being the conditional model density $\rightarrow$ calculation of K-L-divergence between conditional true and model density
- Solution: Scaling the $cAIC$ with the Jacobian J: $JcAIC = -2log(g(\tilde{y}|\hat{\theta}, \hat{u})) - 2log(J(\hat{\lambda}, y)) + 2BC$
- $\tilde{y}$ being the vector of transformed y with transformation $T_\lambda(y)$
- g being the density function of the approximating model
- J being the Jacobian matrix: $J(\hat{\lambda}, y) = \prod_{i=1}^{D} \prod_{j=1}^{N_i} \frac{\partial \tilde{y}_{ij}}{\partial y_{ij}} = \prod_{i=1}^{D} \prod_{j=1}^{N_i} (y_{ij} + s)^{\lambda - 1}$
- BC being a bias correction term, estimated by a bootstrap

Proposed Solution by Lee et al. (2023) II

Consider a Box-Cox transformation ($i = 1, ..., D$ and $j = 1, ..., N_i$):

$$T_\lambda(y_{ij}) = \tilde{y} = \begin{cases} \frac{(y_{ij}+s)^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0 \\ log(y_{ij} + s) & \text{if } \lambda = 0 \end{cases}$$

1. Start with full model with all p variables and estimate $\hat{\lambda}$ for a Box-Cox transformation
2. For each step $s = 1, ..., S$:
   i) Consider all possible models which exclude one predictor
   ii) Estimate $\hat{\lambda}$ for each candidate and calculate $JcAIC$ with estimated $\hat{\lambda}$
   iii) Compare $JcAIC$ values and choose model with lowest value as new optimal model
3. Compare $JcAIC$ from 2.iii) with the previously optimal model from step $s - 1$. Repeat until no better $JcAIC$ can be found.

Methods for the term paper

Methods

- Research question: Does the proposed simultaneous selection and estimation outperform the naive selection in different data scenarios?
- Approach I: Simulation Study
  - Four different data generation processes
  - Four different variable selection mechanisms
  - Inclusion of two irrelevant variables in the model
- Hypotheses
  - "No" and "Log" transformation perform better in Normal and Log settings than the Box-Cox approaches
  - The proposed solution (Box-Cox (Optimal)) comes close to the performance of "No" and "Log"
  - Box-Cox (Optimal) outperforms the naive approach both in terms of $JcAIC$ and variable selection accuracy

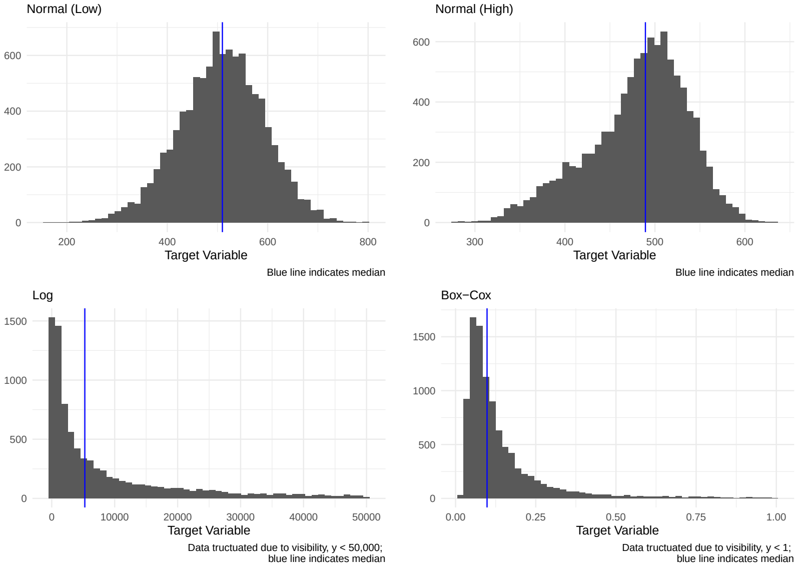
Variable Selection Mechanisms

- "No": $T_\lambda(y_{ij}) = y$ with simple stepwise variable selection on original scale
- "Log": $T_\lambda(y_{ij}) = log(y)$ with simple stepwise variable selection on original scale
- "Box-Cox Naive": First stepwise selection on original scale and Box-Cox transformation afterwards
- "Box-Cox Optimal": Stepwise variable selection using $JcAIC$ on Box-Cox transformed y

Data generating processes

- $x_{1,ij} \sim N(\mu_i, 3^2)$ with $\mu_i \sim U[-3,3]$, $x_{2,ij} \sim Bin(1, 0.8), x_{3,ij} \sim N(0,1)$
- Additional irrelevant variables: $z_1 \sim N(1, 0.1^2)$ and $z_2 \sim Bin(1, 0.4)$
- Normal (Low): $y_{ij} = 400 - 10x_{1,ij} + 100x_{2,ij} - 10x_{3,ij} + u_i + e_{ij}$, $u_i \sim N(0, 30^2), e_{ij} \sim N(0, 60^2)$
- Normal (High): $y_{ij} = 400 - 10x_{1,ij} + 100x_{2,ij} - 10x_{3,ij} + u_i + e_{ij}$, $u_i \sim N(0, 10^2), e_{ij} \sim N(0, 20^2)$
- Log: $y_{ij} = exp\{10 - x_{1,ij} + x_{2,ij} - 0.5x_{3,ij} + u_i + e_{ij}\}$
- Box-Cox: $y_{ij} = [(10 - x_{1,ij} + x_{2,ij} - 0.5x_{3,ij} + u_i + e_{ij}) \cdot (-0.5) + 1]^{\frac{1}{-0.5}}$

Data generating processes

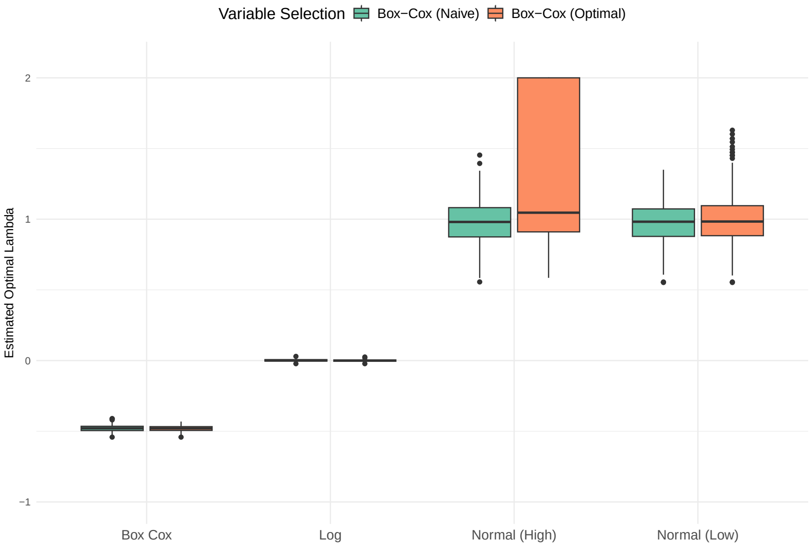


Simulation Setup

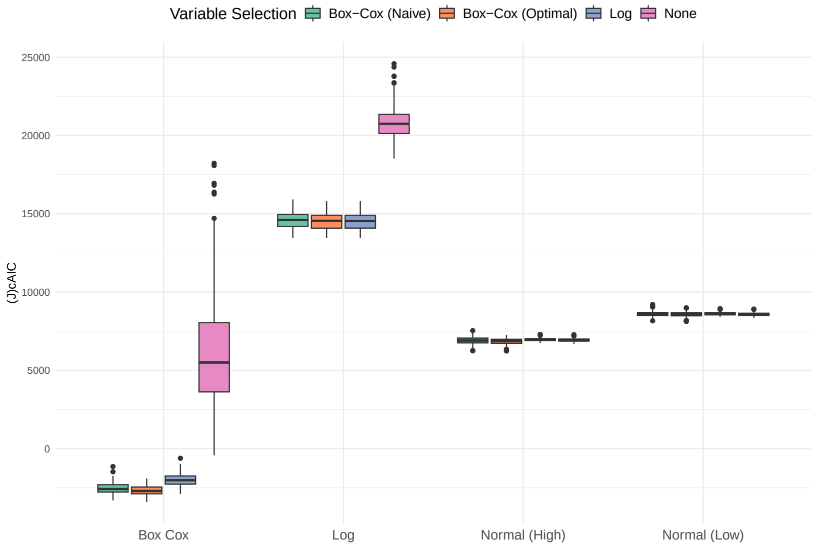
- $R = 300$ iterations
- $N = 10000$ observations in $D = 50$ domains, sampled via random sample ($n_d \sim Unif(0, 30)$)
- Population-generation and random sampling based on sample size in each iteration, but same for all methods
- Interval for estimating λ: $\hat{\lambda} \in \{-2, 2\}$
- Bias correction term estimated via bootstrap with $B = 50$ (paper: $B = 200$, slight differences as will be discussed in the term paper)
- 8 models with Log transformation did not converge in the Normal (High) setup

Results

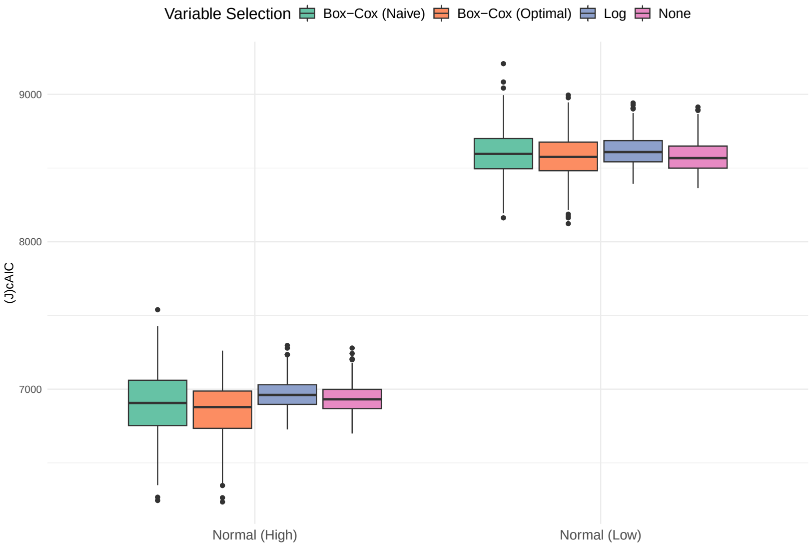
Results: Optimal Lambda's



Results: JcAIC



Results: JcAIC - zoomed into Normal Settings



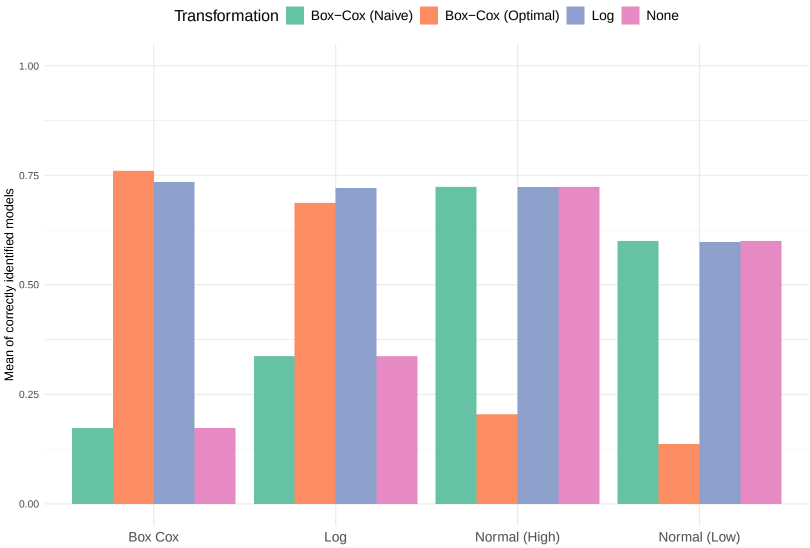
Correctly identified models

Table 1: Percentage of models with lowest JcAIC per iteration and fraction of those which were correct.

DGP	No	Log	Naive	Optimal	Correct
Box Cox	0.00	0.00	0.01	0.99	0.76
Log	0.00	0.52	0.02	0.46	0.69
Normal (High)	0.33	0.01	0.04	0.62	0.37
Normal (Low)	0.50	0.01	0.03	0.46	0.37

- Just half of them has identified the model correctly in the normal setups
- For Log and Box-Cox, high identification rates
- Box-Cox (Optimal) shows lowest $JcAIC$ in almost all cases of Box-Cox and half of the other settings
- Naive selection method does almost not yield lowest $JcAIC$

Results: Correctly identified variables



Zooming In: Normal (High) & Box-Cox (Optimal)

Table 2: Mean and Median Lambda as well as fraction of variables included per correctly identified model or not

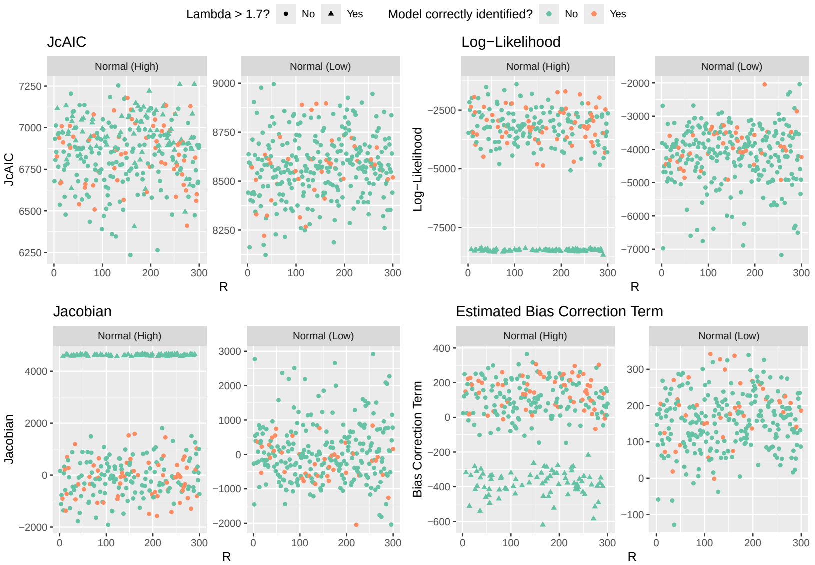
Correct	Mean	Median	SD	x1	x2	x3	z1	z2
No	1.32	1.09	0.50	0.99	0.67	0.78	0.5	0.55
Yes	0.97	0.98	0.16	1.00	1.00	1.00	0.0	0.00

- $\hat{\lambda}$ of false models tends towards 2, for correct models towards 1
- But: $0.9 < \hat{\lambda} < 1.1$ does not ensure correct model specification

Table 3: Mean and Median Lambda as well as fraction of variables included per lambda within [0.9, 1.1]

Lambda	Mean	SD	x1	x2	x3	z1	z2	Correct
0	1.38	0.54	0.99	0.6	0.81	0.37	0.45	0.17
1	0.99	0.05	0.99	1.0	0.86	0.46	0.42	0.26

Closer Look I



Closer Look II

Table 4: Mean of Log-Likelihood, Correction Term, and JcAIC per correctly identified model and estimated lambda, ascending by JcAIC

Model	Lambda	-2*Log-Lik	-2J + 2BC	JcAIC
Correct	> 1.1	8724.23	-1332.82	7391.41
False	> 1.1	13111.84	-5593.94	7530.13
Correct	< 0.9	5471.57	2078.60	7550.16
Correct	Correct	7233.78	354.74	7588.53
False	< 0.9	5801.57	2058.50	7854.05
False	Correct	7571.55	365.24	7899.43

- As reminder:
  $JcAIC = -2log(g(\tilde{y}|\hat{\theta}, \hat{u})) - 2log(J(\hat{\lambda}, y)) + 2BC$
- Correction Term might cancel out worse log-likelihood
- Transformed models seemingly get preferred by the criterion

Discussion

Discussion

- Estimating $\hat{\lambda}$ worked fairly well for both the naive and optimal approach, except of the Normal (High) setup
- Box-Cox (Optimal) often yields models with lowest $JcAIC$
- Variable selection of Box-Cox (Optimal) works (very) well in Box-Cox and Log setting, but not in Normal settings
- Models with correct variables chosen have higher $JcAIC$ than wrong models with transformations
- Next steps:
  - Further investigating path-dependency and skewed distributions
  - Simulating data again for Box-Cox (Optimal) with Normal settings and $B = 200$
  - Mathematical reasoning, esp. for wrong models with high likelihood

Thank you for your attention!

Notes

- RMSE mit original-Daten
- Coverage –> wird noise “rausgerechnet”?

Literature

Lee, Y., Rojas-Perilla, N., Runge, M., & Schmid, T. (2023). Variable selection using conditionalAIC for linear mixed models with data-driven transformations. *Statistics and Computing*,*33*(1), 27.https://doi.org/10.1007/s11222-022-10198-9

Reproduction materials: https://github.com/nippisch/Replication-Lee-et-al.

Appendix

Comparison B = 50 vs. B = 200 with R = 10

dgp	transformation	mean_JcAIC_b50	mean_JcAIC_b200	mean_lambda_b50	mean_lambda_b200	proportion_correct_b50	proportion_correct_b200
boxcox	box.cox	-2785.	-2770.	-0.483	-0.483	0.8	0.9
boxcox	boxcox_naive	-2667.	-2638.	-0.490	-0.489	0.3	0.3
boxcox	log	-2020.	-2010.	NaN	NaN	1.0	1.0
boxcox	no	6470.	6480.	NaN	NaN	0.3	0.3
log	box.cox	14838.	14839.	0.00200	0.00197	0.6	0.8
log	boxcox_naive	14914.	14906.	0.00345	0.00336	0.2	0.2
log	log	14837.	14838.	NaN	NaN	0.6	0.8
log	no	21501.	21511.	NaN	NaN	0.2	0.2
normal_high	box.cox	7046.	7040.	1.26	1.35	0.6	0.6
normal_high	boxcox_naive	7098.	7065.	0.951	0.951	0.8	0.8
normal_high	log	7161.	7173.	NaN	NaN	0.8	0.8
normal_high	no	7126.	7137.	NaN	NaN	0.8	0.8
normal_low	box.cox	8352.	8405.	0.963	0.939	0.3	0.3
normal_low	boxcox_naive	8374.	8410.	0.939	0.939	0.7	0.7
normal_low	log	8481.	8492.	NaN	NaN	0.8	0.7
normal_low	no	8451.	8463.	NaN	NaN	0.7	0.7

Table 5: Comparison of results across B = 50 and B = 200 replications for the Bootstrap of the BC-Term

Results: Optimal Lambda's

- Expectation: $\hat{\lambda} \approx 1$ for Normal settings, $\hat{\lambda} \approx 0$ for Log Setting, and $\hat{\lambda} \approx -0.5$ for Box-Cox-Setting

Table 6: Summary statistics on estimated lambda per DGP and Transformation

DGP	Transformation	Min	Mean	Median	Max
Box Cox	Box-Cox (Naive)	-0.54	-0.48	-0.48	-0.41
Box Cox	Box-Cox (Optimal)	-0.54	-0.48	-0.48	-0.43
Log	Box-Cox (Naive)	-0.02	0.00	0.00	0.03
Log	Box-Cox (Optimal)	-0.02	0.00	0.00	0.03
Normal (High)	Box-Cox (Naive)	0.56	0.98	0.98	1.45
Normal (High)	Box-Cox (Optimal)	0.59	1.25	1.05	2.00
Normal (Low)	Box-Cox (Naive)	0.55	0.98	0.98	1.35
Normal (Low)	Box-Cox (Optimal)	0.55	1.01	0.98	1.63

Results: JcAIC

Table 7: Summary statistics of JcAIC for Normal (Low)

Transformation	Min	Mean	Median	Max
Box-Cox (Naive)	8162.11	8599.32	8596.08	9207.71
Box-Cox (Optimal)	8122.63	8575.04	8575.46	8994.66
Log	8393.09	8617.99	8608.22	8941.14
None	8363.01	8578.25	8567.24	8913.73

Table 8: Summary statistics of JcAIC for Normal (High)

Transformation	Min	Mean	Median	Max
Box-Cox (Naive)	6245.02	6901.62	6906.25	7538.73
Box-Cox (Optimal)	6235.06	6855.44	6878.43	7261.35
Log	6726.79	6963.73	6960.81	7296.72
None	6699.21	6935.16	6931.43	7278.76

Results: JcAIC

Table 9: Summary statistics of JcAIC for Log setting

Transformation	Min	Mean	Median	Max
Box-Cox (Naive)	13455.08	14572.79	14602.64	15916.29
Box-Cox (Optimal)	13454.51	14508.68	14548.80	15798.44
Log	13442.66	14508.01	14535.83	15802.67
None	18531.23	20803.71	20743.94	24583.03

Table 10: Summary statistics of JcAIC for Box-Cox setting

Transformation	Min	Mean	Median	Max
Box-Cox (Naive)	-3305.61	-2542.10	-2577.17	-1139.78
Box-Cox (Optimal)	-3402.87	-2667.26	-2703.47	-1905.03
Log	-2894.39	-1980.86	-2015.43	-611.39
None	-428.27	6183.83	5500.12	18223.78

Results: Correctly identified variables

Table 11: Fraction of variables included in the models for Normal (Low) Setup

Transformation	x1	x2	x3	z1	z2	Correct
Box-Cox (Naive)	1.00	1.00	0.78	0.14	0.13	0.60
Box-Cox (Optimal)	0.82	0.93	0.59	0.39	0.42	0.14
Log	1.00	1.00	0.79	0.14	0.13	0.60
None	1.00	1.00	0.78	0.14	0.13	0.60

Table 12: Fraction of variables included in the models for Normal (High) Setup

Transformation	x1	x2	x3	z1	z2	Correct
Box-Cox (Naive)	1.00	1.00	0.99	0.16	0.14	0.72
Box-Cox (Optimal)	0.99	0.73	0.83	0.40	0.44	0.20
Log	1.00	1.00	0.99	0.17	0.14	0.72
None	1.00	1.00	0.99	0.16	0.14	0.72

Results: Correctly identified variables

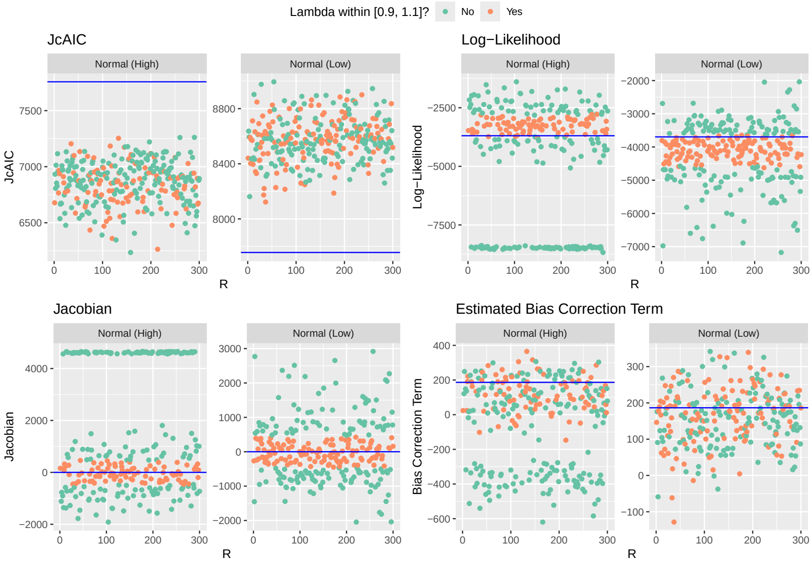
Table 13: Fraction of variables included in the models for Log Setup

Transformation	x1	x2	x3	z1	z2	Correct
Box-Cox (Naive)	1	0.55	0.84	0.16	0.13	0.34
Box-Cox (Optimal)	1	1.00	0.99	0.16	0.17	0.69
Log	1	1.00	0.99	0.15	0.15	0.72
None	1	0.55	0.84	0.16	0.13	0.34

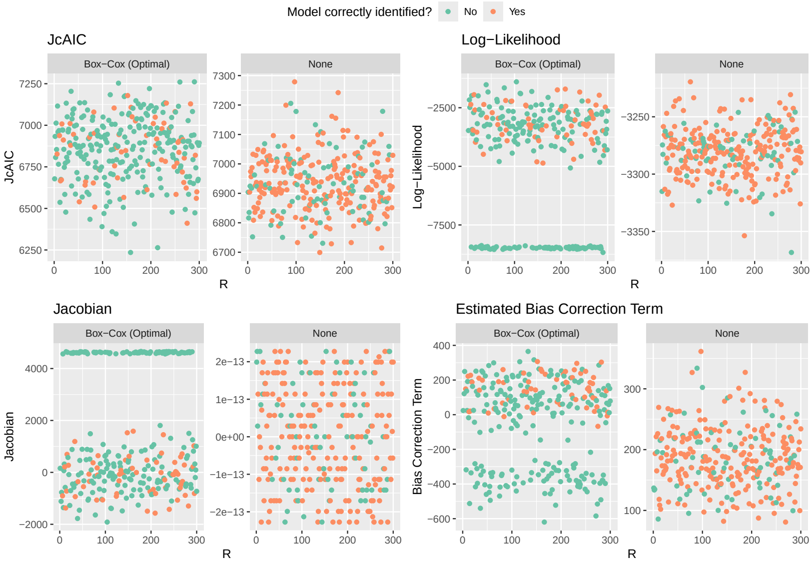
Table 14: Fraction of variables included in the models for Box-Cox Setup

Transformation	x1	x2	x3	z1	z2	Correct
Box-Cox (Naive)	0.99	0.53	0.42	0.16	0.10	0.17
Box-Cox (Optimal)	1.00	0.99	1.00	0.11	0.14	0.76
Log	1.00	1.00	1.00	0.14	0.15	0.73
None	0.99	0.53	0.42	0.16	0.10	0.17

Closer Look II



Comparison with no transformation



Not necessarily just skewed densities